

LazyBatchNorm2d#

[https://pytorch.org/docs/stable/generated/torch.nn.LazyBatchNorm2d.html#torch](https://pytorch.org/docs/stable/generated/torch.nn.LazyBatchNorm2d.html#torch.nn.LazyBatchNorm2d)

Description:

A `torch.nn.BatchNorm2d` module with lazy initialization. Lazy initialization is done for the `num_features` argument of the `BatchNorm2d` that is inferred from the `input.size(1)`. The attributes that will be lazily initialized are `weight`, `bias`, `running_mean` and `running_var`. Check the `torch.nn.modules.lazy.LazyModuleMixin` for further documentation on lazy modules and their limitations. `eps` (float) – a value added to the denominator for numerical stability. Default: `1e-5` `momentum` (Optional[float]) – the value used for the `running_mean` and `running_var` computation. Can be set to `None` for cumulative moving average (i.e. simple average). Default: `0.1`

Function Signature

```
class torch.nn.LazyBatchNorm2d(eps=1e-05, momentum=0.1,  
affine=True, track_running_stats=True, device=None,  
dtype=None)[source]#
```

Parameters

```
cls_to_become[source]#
```

Detailed Documentation

Documentation

A `torch.nn.BatchNorm2d` module with lazy initialization.

Lazy initialization is done for the `num_features` argument of the `BatchNorm2d` that is inferred from the `input.size(1)`. The attributes that will be lazily initialized are `weight`, `bias`, `running_mean` and `running_var`.

Check the `torch.nn.modules.lazy.LazyModuleMixin` for further documentation on lazy modules and their limitations.

`eps` (float) – a value added to the denominator for numerical stability. Default: $1e-5$

`momentum` (Optional[float]) – the value used for the `running_mean` and `running_var` computation. Can be set to `None` for cumulative moving average (i.e. simple average). Default: 0.1

`affine` (bool) – a boolean value that when set to `True`, this module has learnable affine parameters. Default: `True`

`track_running_stats` (bool) – a boolean value that when set to `True`, this module tracks the running mean and variance, and when set to `False`, this module does not track such statistics, and initializes statistics buffers `running_mean` and `running_var` as `None`. When these buffers are `None`, this module always uses batch statistics. in both training and eval modes. Default: `True`

